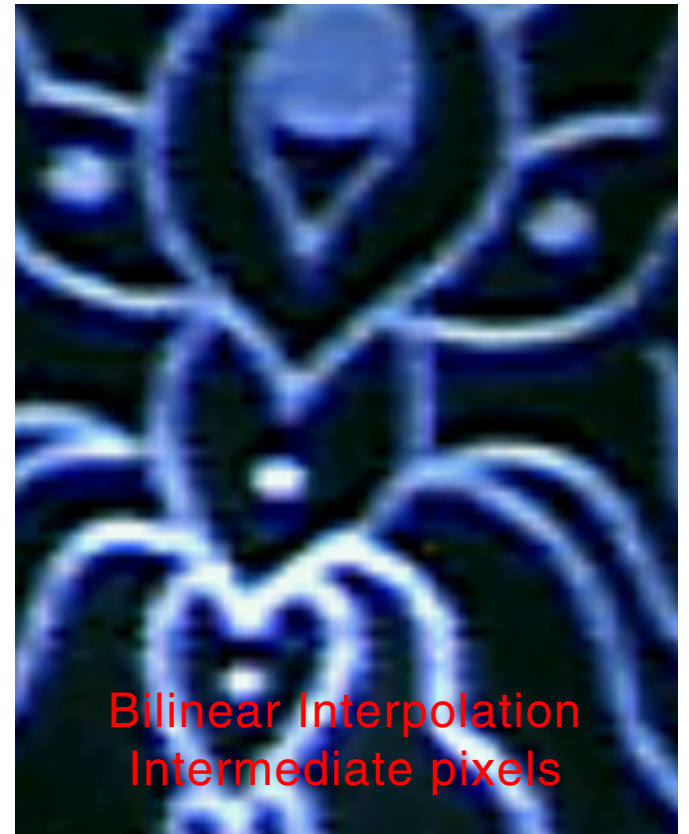


Interpolation for Scaling, Rotation, Perspective and Morphing

Between two neighbour pixels the distance is 1. Using scalefactor $S=8$ means: 7 pixels must be inserted in each direction. This is done by pixel-doubling, bilinear, biquadratic or bicubic interpolation. Using the zoom shows that indeed one left box-like pixel is substituted by 64 pixels in the right image.



Best view
Zoom 100%



Interpolation Principles (Images on next Page)

The green plane represents 4x4 pixels of an original RGB image. The image itself is not illustrated.

The vertical stakes show the values of one of the three RGB colors, e.g. Green.

The stakes are covered by a smooth surface like an elastic skin.

For biquadratic or bicubic Interpolation the surface is parabolic of second or third order.

For bilinear Interpolation, the stakes are connected by twisted planes. The roof looks then polygonal instead of smooth. This is for many cases good enough, mainly if the interpolated image doesn't have considerably more pixels.

The image shall have a different number of pixels, but the geometrical size is the same.

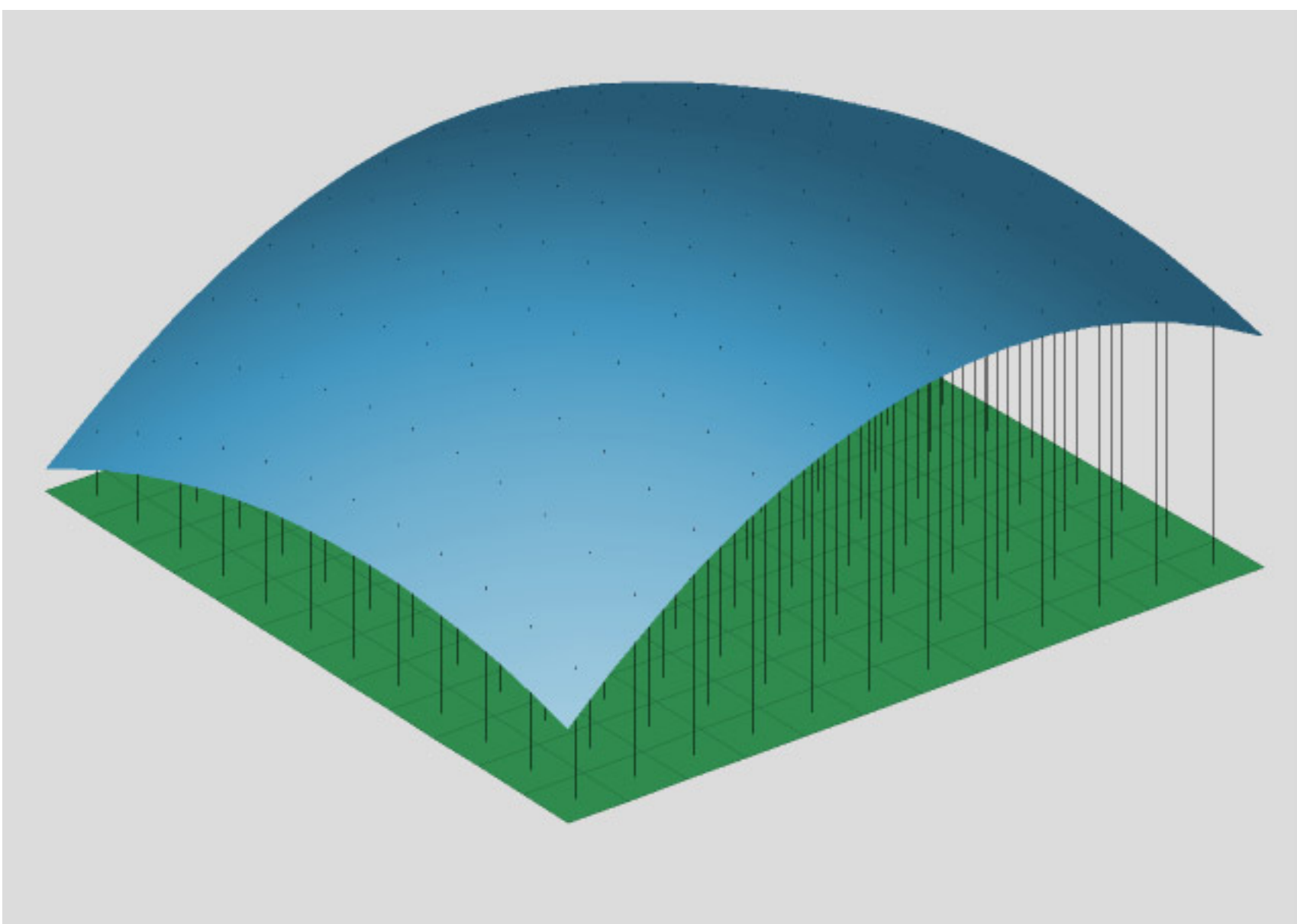
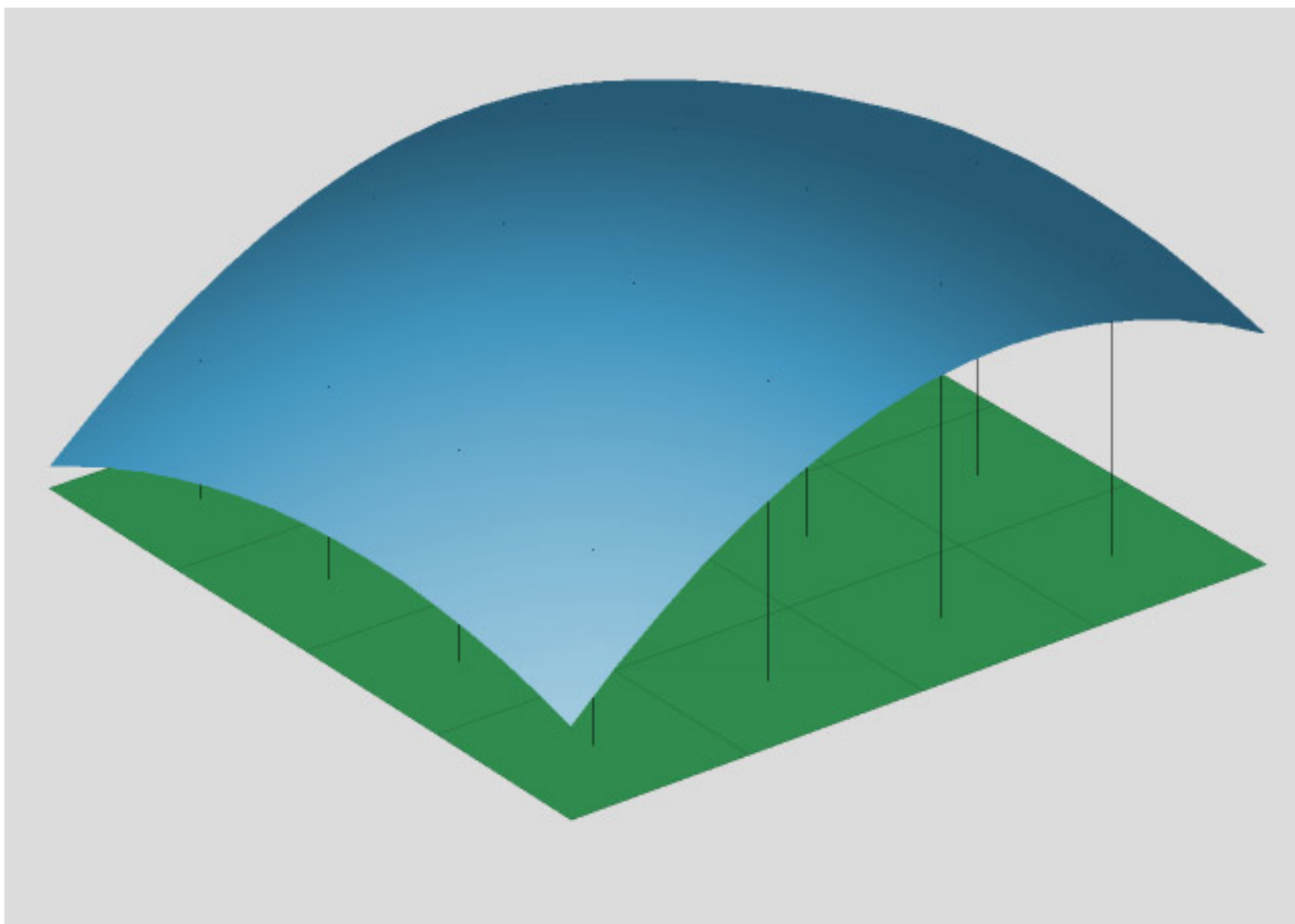
The green plane represents now 12x12 pixels for the image.

Intermediate pixels can be computed by taking the values for e.g. Green from the smooth surface.

The example shows upsampling by factor $S=3$. Any factor is possible, e.g. $S=2.345$.

This method has to be applied for scaling, rotation, perspective transformations and morphing.

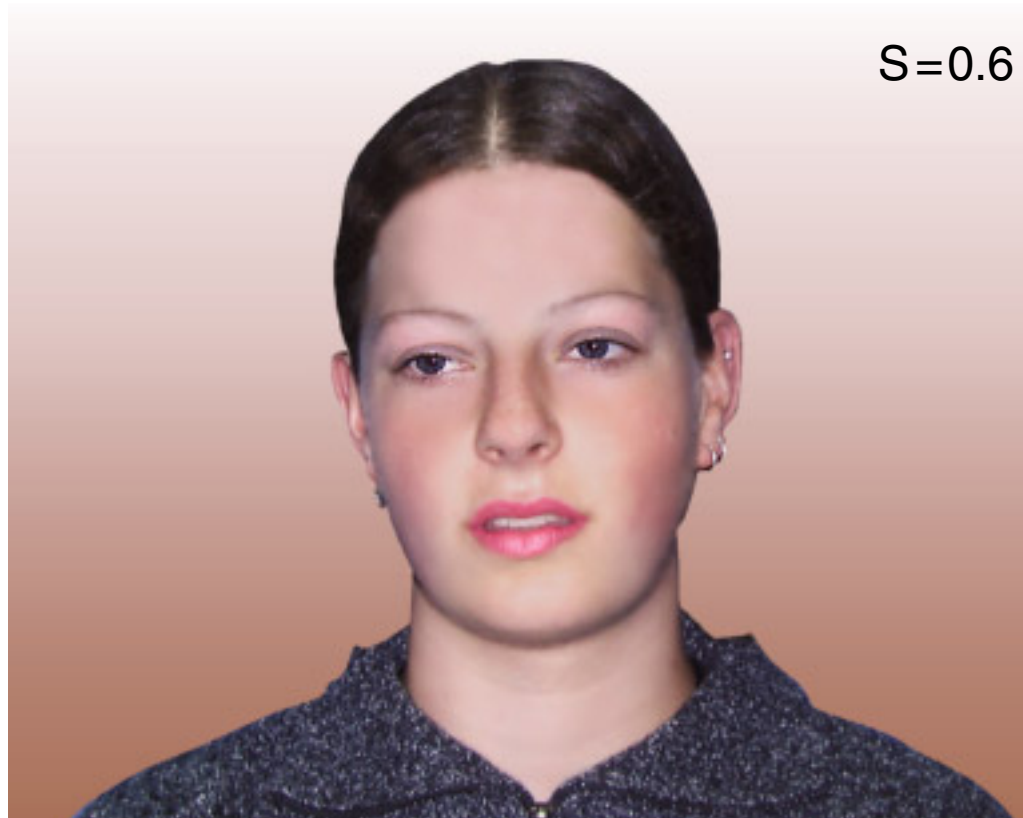
Interpolation for Upsampling



Interpolation for Downsampling

ZEBRA uses a linear interpolator. For downsampling this interpolator works as an averager. This is accurate for scalefactors down to about $S=0.5$.

Downsampling for e.g. $S=0.3$ is done by downscaling first by $S=0.6$ and then in a second step by $S=0.5$. The averager is applied twice.



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